ClimateResilience-India: final report

Generated 2026-10-16 by `scripts/make_report.py`. Every number below is read from `reports/*.json` and `reports/*.csv`; a number that could not be found is printed as `MISSING (...)`.

1. Summary

ClimateResilience-India is a district-level decision-support system for four southern states (Andhra Pradesh, Telangana, Karnataka, Tamil Nadu). It ingests weather and crop-yield data, stores them in a PostGIS warehouse, analyses climate risk, forecasts yields, warns about heatwaves and droughts, optimizes irrigation, shows the season in a 3D simulator and delivers advice through Streamlit dashboards and a WhatsApp/SMS assistant.

What the measurements say, plainly:

Area	Measured result (held-out years)	What it means
Data	51 district units, 633,369 daily weather rows, 4,286 yield records; satellite table 100% synthetic	all data-quality checks passed; satellite features are not used by the served models
Yield forecast (2015-17, 60-day lead)	relative RMSE 40.4% (pooled RMSE 1.04 t/ha, R^2 0.91); best simple baseline (district trend) 40.9%	Level with the best baselines: ratio vs district trend 0.99 (95% interval 0.94 to 1.03); clearly better than last year's yield (0.81). 80% interval covers 80% of test cases
Drought warning (2020-24)	F1 0.50, PR-AUC 0.38 vs climatology 0.09 (base rate 7.5%)	Real skill over climatology and persistence (0.13); validation F1 was 0.70
Heatwave warning (14 days)	LSTM PR-AUC 0.014 vs climatology 0.064 on 37 pooled onsets	No skill over climatology, so the **climatology** probability is served
Irrigation optimizer (normal rain)	saves 61% of the 600 mm baseline water; keeps 98% of baseline yield	Modelled, not field-tested. Vs an even split: 161 mm less water, yield difference +0.013 (interval +0.0005 to +0.0268), worse in 7 of 20 cases
Dashboards	7 addresses, 0 failed browser checks, median warm page load 1.6-2.6 s	Headless Chromium with software rendering
Advisory bot	7 of 7 smoke checks passed	Dry run by default; translations unreviewed
Tests	336 automated tests	`make test`; in a clean Linux container: 327 passed, 9 skipped in 71.21s (0:01:11)

- The yield models are **not better than simple baselines** by a margin the data can confirm. They are useful as a weather-adjusted baseline with an honest interval, not as precise predictors.
- The heatwave LSTM has **no skill over climatology**; the system serves climatology and says so. The drought model **does** beat climatology and persistence.
- The optimizer **saves a large share of the water but keeps less yield than the 600 mm baseline** in the model. Against an even split of the same kind of budget it uses less water with a clear but small yield gain. The plan's story that yield recovers while water is saved is not what the models show.
- Satellite NDVI/soil-moisture inputs are **synthetic** (Earth Engine was not authenticated). The served models do not use them.

2. Data and scope

Units are the 51 yield-reporting district groups that ICRISAT publishes (1966 boundaries, reconstructed from 2021 polygons), not villages. Sources: IMD gridded rainfall and temperature, NASA POWER (humidity, wind, solar), ICRISAT district yields (2000-2017), geoBoundaries polygons. Rows: 633,369 daily weather, 4,286 yield records, 29,325 satellite rows (100% synthetic, flagged `is_synthetic`). Data-quality verdict: all data-quality checks passed (`docs/DATA_QUALITY.md`). Decisions that shape the results are logged in `docs/DECISIONS.md`.

3. Climate findings (descriptive and diagnostic)

1. **Rainfall variability differs sharply between districts.** The coefficient of variation of kharif rainfall averages 0.28 across 51 districts (1991-2024), against 0.71 for rabi. The most variable district is Thirunelveli (CV 0.59, mean 321 mm); the most stable is Dakshina Kannada (CV 0.14). Across districts, mean rainfall and CV have Spearman rho = -0.76 ($p < 0.001$). The driest regional years were 2002 (100% of districts below normal), 2003 (80% of districts below normal), 2023 (80% of districts below normal).
2. **Yield trends differ by crop and state.** Of 13 state-crop series (1991-2017, balanced district panels), 8 show a significant increase and 1 a significant decrease (Mann-Kendall, $p < 0.05$). Rice rises significantly in 3 of 4 states (Andhra Pradesh +0.045 t/ha/yr, +1.5%/yr, Karnataka +0.023 t/ha/yr, +0.9%/yr, Telangana +0.045 t/ha/yr, +1.6%/yr); the steepest relative gain is cotton in Telangana (+3.2% of mean per year).
3. **Kharif droughts (season rain < 75% of the district mean) hit about one season in 6.** Mean frequency is 18.1% of 34 seasons; Thirunelveli is most exposed (11 events, 32%). The worst regional years were 2002 (73% of districts), 2016 (53% of districts), 2003 (49% of districts). Under the strict flood-risk definition (> 125% of mean and >= 3 days > 100 mm of district-mean rain) there were only 11 events in 9 districts, because district-area-mean rainfall rarely exceeds 100 mm/day.
4. **Pooled and within-district views of weather and yield can disagree.** Pooled across districts, rice yield and kharif rainfall have Pearson r = -0.17 (95% cluster-bootstrap CI -0.33 to -0.02); within districts, after removing each district's level and trend, r = 0.19 (0.11 to 0.28). The sign flips: wetter districts yield less rice, but a wetter year helps a given district (spatial confounding, so pooled correlations are misleading). In the two-way fixed-effects regression (district and year effects, district-clustered SEs, n = 918), +100 mm of rainfall is associated with higher yield (+0.024 t/ha, $p < 0.001$) and +1 C of mean Tmax has no statistically detectable association with yield (-0.107 t/ha, $p = 0.070$); in the district-FE-plus-trend model Tmax is -0.147 t/ha ($p < 0.001$) and rain +0.022 t/ha ($p = 0.003$). Rain and Tmax together add 0.6% of R-squared beyond district and year effects, which sets a low ceiling for any weather-based yield forecast.
5. **Districts fall into 7 climate groups** (KMeans, silhouette 0.37, features from years <= 2011 only): dry-hot-variable-rain-long-dry-spells (3 districts, mean kharif rain 431 mm, e.g. Madurai, Thirunelveli); dry-hot-variable-rain-long-dry-spells-volatile-yield (2 districts, mean kharif rain 456 mm, e.g. Ramananthapuram, Thanjavur); moderate-warm (15 districts, mean kharif rain 545 mm, e.g. Ananthapur, Bellary); moderate-warm-stable-rain (7 districts, mean kharif rain 746 mm, e.g. Belgaum, Bidar); moderate-hot (13 districts, mean kharif rain 834 mm, e.g. Adilabad, East Godavari); moderate-cool (9 districts, mean kharif rain 880 mm, e.g. Bangalore, Chickmagalur); wet-cool (2 districts, mean kharif rain 3155 mm, e.g. Dakshina Kannada, Uttara Kannada).

Full tables: `docs/FINDINGS.md`, `reports/analytics/`.

4. Yield forecasting

Target: yield relative to the district's prior mean. Time-based splits only (train to 2011, validation 2012-14, test 2015-17). Models: naive last year, district trend, prior-three-year mean, Ridge, random forest, XGBoost (Optuna, monotone temperature constraint), LSTM; 80% intervals by conformalized quantile regression.

- Test relative RMSE of the served XGBoost: 40.4%; best simple baseline (district trend): 40.9%. In absolute terms the pooled RMSE is 1.04 t/ha against 1.27 t/ha for that baseline; the best baseline by pooled RMSE is the prior mean 3y (0.97 t/ha, lower than the model's). Pooled errors are dominated by sugarcane's high yields, and R^2 (0.91) says little: last year's yield alone already reaches 0.88, because yields differ strongly between districts.
- Cluster-bootstrap ratios of relative RMSE (below 1 is better): vs last year 0.81; vs district trend 0.99 (0.94 to 1.03); vs prior three-year mean 0.96 (0.92 to 1.00).
- 80% interval coverage on the test years: 80%.
- What-if sensitivity of the served model (validation years): -2.2% for 40% less rain, -3.5% for +4 °C: the model learned yield mostly from each district's own history.
- Full tables and SHAP: `reports/yield_model_comparison.md`, `reports/yield_metrics.csv`, `reports/figures/yield_shap_*.png`.

5. Heatwave and drought early warning

- **Drought** (SPI-3 at or below -1 next month; XGBoost on rainfall-deficit features, isotonic calibration): test F1 0.50 (precision 0.41, recall 0.64), PR-AUC 0.38 against 0.09 for climatology and 0.13 for persistence; validation F1 0.70. A drought signal for a normally dry month does not set the risk level (D-024).
- **Heatwave** (onset within 14 days, March-June): pooled rolling-origin PR-AUC 0.014 for the LSTM against 0.064 for climatology over 37 onsets; the served probability is **climatology**. Onsets are rare, so conclusions are weak in both directions.
- Details: `reports/extreme_weather_report.md`.

6. Irrigation optimization

Daily FAO-56 water balance, FAO-33 yield response and a weekly linear programme (PuLP/CBC); the slowest LP case took 0.12 s. Paddy rice is modelled as a ponded field (standing water behind bunds and seepage while it stands; the parameters are approximate and unverified, D-029); the other crops use the plain root-zone balance.

- Normal rainfall: 61% water saved against the 600 mm baseline, yield 98% of baseline. Dry year (rain x0.7): 56% saved, yield 96% of baseline.
- Against an even split: 161 mm less water, relative-yield difference +0.013 (interval +0.0005 to +0.0268), worse in 7 of 20 cases.
- Crop parameters and prices are approximate and unverified (D-021, D-022). Results are model output, not field trials.

7. Simulator, dashboards, advisory

- **3D simulator** (Three.js, one self-contained page): crop-specific plant models that grow over the season, flooded and terraced paddies, soil that darkens when wet and cracks when dry, daily weather with dates, a time-of-day control, an uncertainty view and a by-stage stress panel; the relief is real (open terrain tiles) but fields, plants and soil classes are schematic. Checked in headless Chromium with software rendering (6 districts, 2772 scenarios), which ran at 9 frames per second; software rendering is far slower than a GPU, so this is a lower bound, not a measurement of the 30 fps goal. Frame rate on real hardware: MISSING (not measured: needs a real browser on real hardware).
- **Dashboards** (Streamlit, a home page and six pages): 7 addresses loaded in a browser with 0 failed checks (no exception boxes, no failed requests); median warm page loads 1.6-2.6 s.
- **Advisory:** fixed templates in English, Hindi, Tamil and Telugu, at most 320 characters; WhatsApp webhook with signature validation; sending is a dry run unless explicitly switched; 7 of 7 bot smoke checks passed. Translation review: MISSING (no native-speaker review done).
- **Container:** 7 of 7 checks passed (serving image 3,525 MB, non-root, health check answers, models run from the demo bundle, memory 172.5MiB / 8.729GiB); the full test suite inside a container built from `requirements.lock`: 327 passed, 9 skipped in 71.21s (0:01:11) (browser tests skip there: no Chromium or node in the image).
- **Automated tests:** 336 (`make test`), plus the browser checks (`make test-sim test-dashboards test-bot`).
- **Deployed URL:** MISSING (deployment is a manual step, see docs/DEPLOY.md).

8. The plan's targets against what was measured

Plan hypothesis	Measured	Verdict
Yield RMSE 2.9 and R^2 0.82	pooled RMSE 1.04 t/ha, R^2 0.91, relative RMSE 40.4%	Not comparable as stated (the plan fixes neither units nor pooling); the honest comparison is against baselines, where there is no clear gain
F1 0.75 for extreme weather	drought F1 0.50 (test), 0.70 (validation); heatwave served as climatology	Not reached on held-out years
30% water saving	61% (normal), 56% (dry) against the 600 mm baseline	Exceeded in the model, at 98% / 96% of baseline yield

The plan's figures were hypotheses, not results; they are shown here so that nobody mistakes them for achievements.

9. Limitations

- Yields end in 2017; nothing is validated on recent seasons. There is no live weather feed: early-warning pages replay observed history.
- District units are reconstructed 1966 parents; results are not village-level.
- Satellite features are synthetic; weather is IMD 1-degree temperature and 0.25-degree rain grids averaged over polygons.
- Translations, advice wording, crop parameters and prices are unreviewed or unverified. Data licences for redistribution (IMD, ICRISAT, geoBoundaries ODbL) must be checked before publishing the

demo bundle (`scripts/check_licenses.md`).

10. Reproduce

```
 make setup && make db-up
make pipeline            # etl -> warehouse -> analytics -> models -> optimization -> simulator (ho
make test                # unit tests;  make test-sim test-dashboards test-bot  for the browser and
python scripts/make_report.py
```

Without the raw data: `python -m config.bundle restore` copies the demo bundle (`data/sample/`) into place and `make dashboards` runs the app.

11. Attribution

IMD (gridded rainfall and temperature); NASA Langley Research Center POWER Project; ICRISAT/TCI District Level Database; geoBoundaries (Runfola et al. 2020) with its source data (ODbL); three.js (MIT). See `scripts/check_licenses.md`.